Exploratory Data Analysis and Comparative Machine Learning for Air Quality Data

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Predictive Analytics Course Project

Abstract- This project presents a data-cleaning, exploratory data analysis (EDA), and machine-learning workflow for an air-quality dataset containing pollutant measurements. The initial data showed 436 missing values in each of the pollutant_min, pollutant_max, and pollutant_avg fields. Rows without usable pollutant values were removed, and no duplicate rows were found. The cleaned dataset contains 3,372 records and 11 columns. EDA was used to examine the distribution of pollutant averages, potential outliers, correlations, and the relationship between minimum and maximum pollutant measurements. Five supervised models were included: Linear Regression and Polynomial Regression for numeric estimation, and Logistic Regression, K-nearest neighbors (KNN), and support vector machine (SVM) for classification. Polynomial Regression achieved the stronger regression result (R2 = 0.9564), while Logistic Regression achieved 95.85% test accuracy and KNN achieved 95.41%. The SVM result shown in the supplied visualization is 55.40% training accuracy and is not directly comparable with the held-out results. The findings also require care because the minimum and maximum pollutant measurements are strongly associated with the average value used as the target.

Keywords- air quality, pollutant average, exploratory data analysis, data preprocessing, regression, classification, KNN, SVM

I. INTRODUCTION

Air-quality measurements can be examined to understand the range and variation of pollutant levels across the observations in a dataset. Before applying machine-learning algorithms, the data should be checked for missing values, duplicate records, unusual values, and relationships among numerical features. These checks help describe the dataset and provide context for interpreting model performance.

This project uses pollutant_min, pollutant_max, and pollutant_avg as its main measurement fields, together with latitude and longitude as location-related features. The work has three objectives: (1) clean and summarize the available records, (2) use EDA charts to examine pollutant distributions and relationships, and (3) compare two regression models and three classification models. Regression estimates a numerical pollutant average, while classification separates records into two groups using the median pollutant average as the threshold.

The classification labels used in this project are created from the dataset median. They are project-defined categories and should not be interpreted as official AQI bands or health categories. The analysis is also an estimation task using fields from the same observation; it does not, by itself, establish that future pollution levels can be forecast.

II. DATASET AND PREPROCESSING

A. Dataset Description

The CSV dataset is titled Real time Air Quality Index from various locations and is available through the Open Government Data (OGD) Platform India. It is published by the Central Pollution Control Board (CPCB), under the Ministry of Environment, Forest and Climate Change. The supplied file contains 11 columns: country, state, city, station, last_update, latitude, longitude, pollutant_id, pollutant_min, pollutant_max, and pollutant_avg. The numerical fields used directly in this analysis are latitude, longitude, pollutant_min, pollutant_max, and pollutant_avg. The last field is used as the regression target and to create the binary classification label [1]. The measurement units are not specified in the supplied CSV.

B. Data Cleaning

The initial missing-value check reported 436 missing entries in each of pollutant_min, pollutant_max, and pollutant_avg. Rows with missing values in these three required pollutant fields were removed using dropna. The duplicate-row check returned zero duplicates, after which drop_duplicates was applied. The resulting cleaned dataset contains 3,372 rows and 11 columns. The cleaned file was saved as AQI_Preprocessed.csv.

Removing rows without the pollutant values required for modelling allows the regression and classification algorithms to use complete target and predictor values. No duplicate observations were identified in the supplied data. The same cleaned dataset was then used for the EDA and modelling stages to keep the analysis consistent.

C. Target Preparation

For regression, pollutant_avg is treated as a continuous numerical target. For classification, a binary target is created by comparing pollutant_avg with its median: records at or below the median are assigned class 0, and records above the median are assigned class 1. Logistic Regression and KNN use an 80:20 train-test split with random_state=42. Feature scaling is applied to the KNN training data and then to the test data using the fitted scaler.

III. EXPLORATORY DATA ANALYSIS

The histogram of pollutant_avg shows a right-skewed distribution. Most observations are concentrated at lower pollutant-average values, while a smaller number extend to much higher values, with the visible tail reaching about 200. This indicates that the distribution is not symmetric and that a limited number of high observations may influence summary statistics and prediction errors.

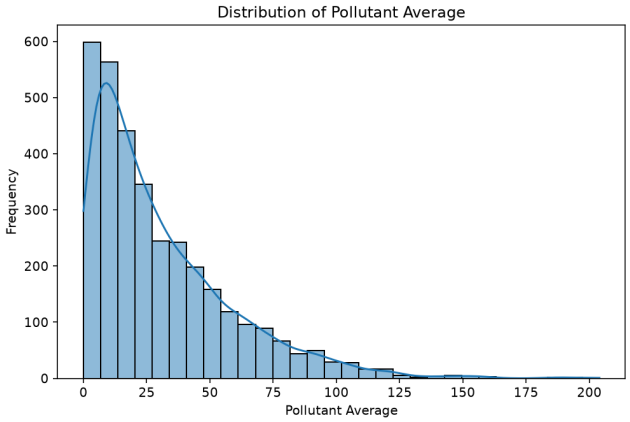


Fig. 1. Distribution of pollutant average.

The boxplot also shows a long upper tail and many observations beyond the upper whisker. Several high values extend beyond 150 and a few are close to 200. These points are potential outliers, but the plot alone does not prove that they are incorrect measurements. They are therefore described rather than automatically removed.

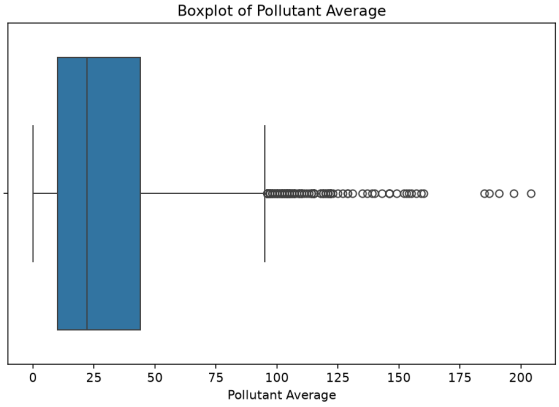


Fig. 2. Boxplot of pollutant average showing high-value outliers.

The correlation heatmap shows a strong positive relationship between pollutant_avg and pollutant_max (0.85), and between pollutant_avg and pollutant_min (0.84). The correlation between pollutant_min and pollutant_max is 0.53. Latitude has a very small correlation with pollutant_avg (0.01), while longitude has a weak negative correlation (-0.17). These are linear correlations across the full dataset and do not establish causation.

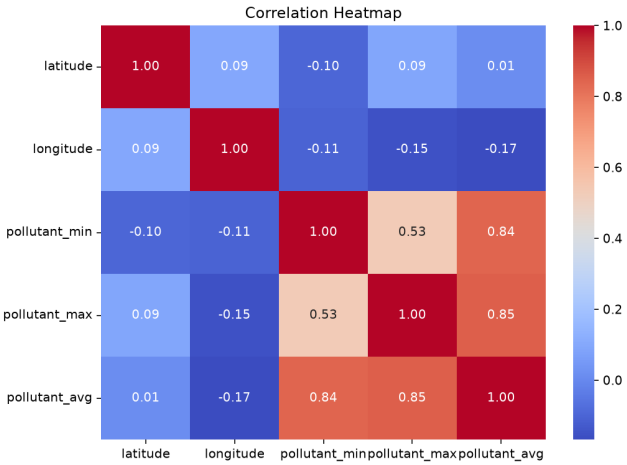


Fig. 3. Correlation heatmap for the five numerical features.

The scatterplot of pollutant_min against pollutant_max shows an overall positive pattern: larger minimum values are generally associated with larger maximum values. The spread of points also increases at higher values, and some observations sit well above the main cluster. This agrees with the positive correlation in the heatmap.

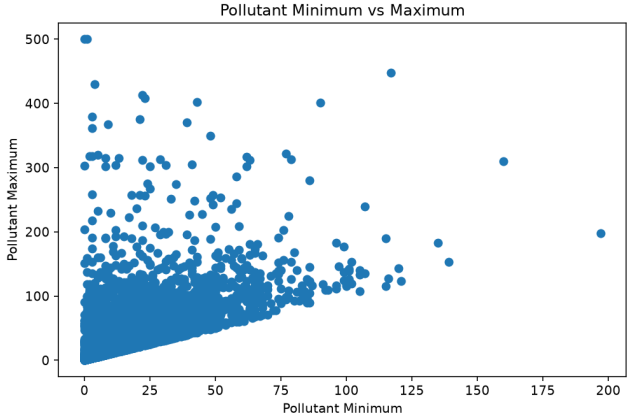


Fig. 4. Relationship between pollutant minimum and maximum.

IV. MACHINE-LEARNING METHODOLOGY

A. Regression Models

The regression target is pollutant_avg. Linear Regression uses latitude, longitude, pollutant_min, and pollutant_max as input features. Polynomial Regression first expands the same features using degree-two polynomial terms and then fits a Linear Regression estimator. The regression models use an 80:20 train-test split with random_state=42. Their performance is measured using mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), and the coefficient of determination (R2). Lower MAE, MSE, and RMSE indicate smaller errors, while a higher R2 indicates that the model explains more variation in the held-out target.

B. Classification Models

Logistic Regression and KNN use the binary median-based target and the four numerical input features. The KNN features are standardized before fitting because the method uses distances between observations. The SVM visualization supplied for this project uses latitude and longitude with a linear kernel to display a two-feature decision boundary. Its printed accuracy of 55.40% is calculated on the same data used to fit that SVM model. It is therefore training accuracy, not a held-out test score, and should not be compared directly with the Logistic Regression and KNN test accuracies.

For binary classification, accuracy is the proportion of correctly classified test records. A confusion matrix shows correct and incorrect predictions for each class. Precision measures the correctness of positive predictions, recall measures how many actual positive records are found, and the F1-score balances precision and recall. The comparison in this report emphasizes the available accuracy values and the supplied confusion-matrix and prediction-performance figures.

V. RESULTS AND DISCUSSION

A. Regression Results

Table I summarizes the regression metrics obtained from the project code. Polynomial Regression has lower MAE, MSE, and RMSE than Linear Regression, and its R2 is higher. Based on these values, Polynomial Regression performs better on the held-out regression split.

Model	MAE	MSE	RMSE	R2
Linear Regression	4.0228	50.2924	7.0917	0.9343
Polynomial Regression	2.9972	33.3523	5.7751	0.9564

TABLE I. REGRESSION PERFORMANCE ON THE TEST SET

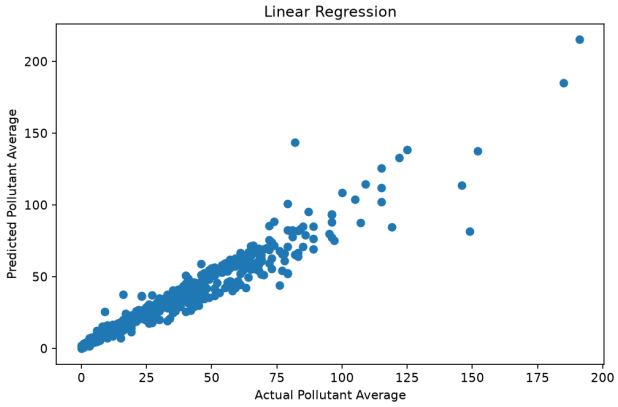


Fig. 5. Actual versus predicted values for Linear Regression.

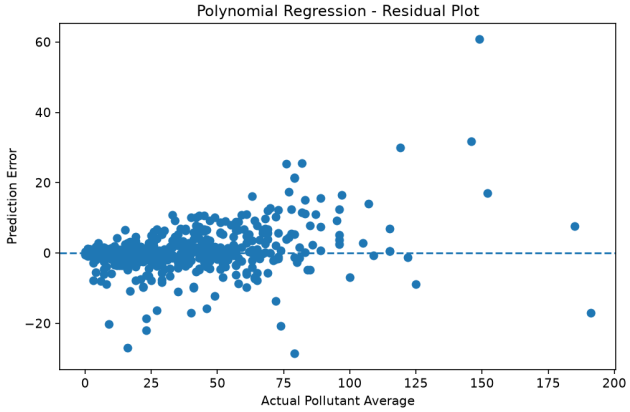


Fig. 6. Residual plot for Polynomial Regression.

The actual-versus-predicted plot shows that many predictions follow the general increase in actual pollutant average, although the spread becomes larger at higher values. The polynomial residual plot also shows a wider range of errors for some larger actual values. This suggests that high-value observations remain more difficult to estimate accurately even though the overall regression scores are high.

B. Classification Results

Logistic Regression achieved 95.85% accuracy on its test set, while KNN achieved 95.41%. Their results are close, with Logistic Regression slightly higher in the initial evaluation. The Logistic Regression confusion matrix contains 318 correctly classified class-0 records and 329 correctly classified class-1 records. It also contains 13 false positives and 15 false negatives.

Classifier	Accuracy	Evaluation
Logistic Regression	95.85%	Test set
KNN (k=5)	95.41%	Test set
Linear SVM	55.40%	Training set

TABLE II. CLASSIFICATION ACCURACY REPORTED BY THE PROJECT CODE

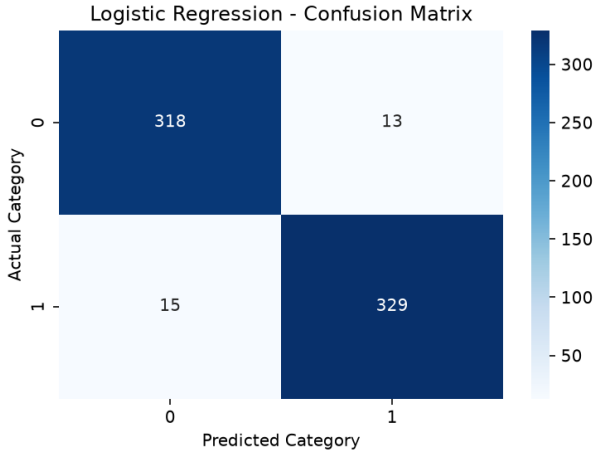


Fig. 7. Logistic Regression confusion matrix.

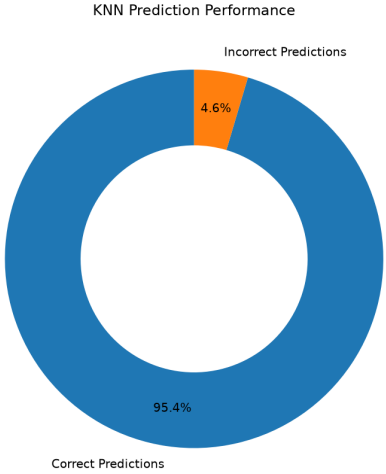


Fig. 8. KNN correct and incorrect prediction proportions.

The KNN performance chart shows approximately 95.4% correct predictions and 4.6% incorrect predictions. The Logistic Regression and KNN results suggest that the median-based binary classes can be separated well using the selected features on the current random test split. However, the label is derived from pollutant_avg itself and is not an independently supplied AQI category.

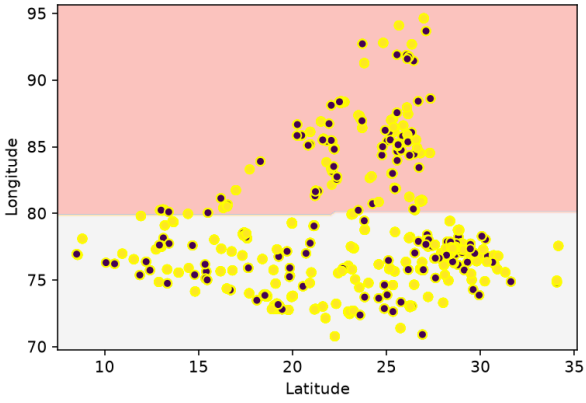


Fig. 9. Linear SVM decision-boundary visualization using latitude and longitude.

The SVM decision-boundary figure provides a two-dimensional view using latitude and longitude. Its displayed 55.40% accuracy is measured on the training observations, and the SVM uses a different feature set from the other classifiers. A fair model ranking requires retraining SVM with the same four features and evaluating it on the same held-out split used for Logistic Regression and KNN.

C. Model Comparison

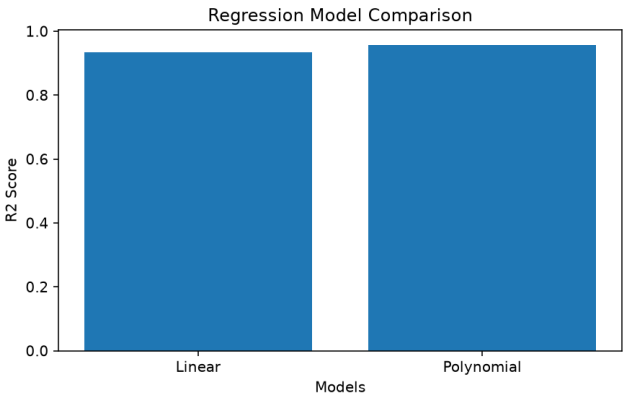


Fig. 10. Regression model comparison using R2.

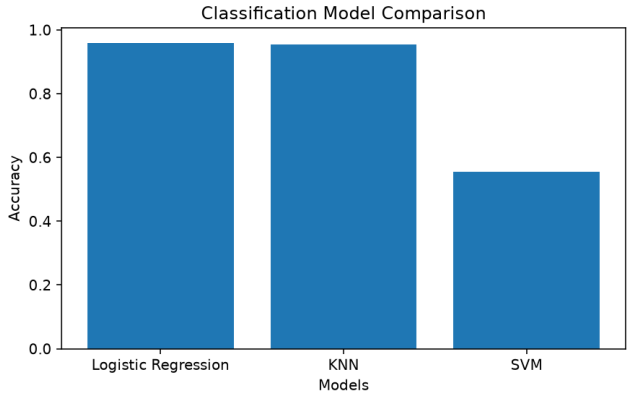


Fig. 11. Classification accuracy comparison; SVM bar is training accuracy.

The regression comparison chart confirms the higher R2 obtained by Polynomial Regression. The classification chart shows Logistic Regression and KNN close to one another and the SVM result lower. Since the SVM value is training accuracy while the other values are test accuracies, the chart is a visual summary of the values currently available rather than a fully controlled comparison. A new SVM test result is needed before choosing the best classifier conclusively.

VI. LIMITATIONS AND FUTURE WORK

Several limitations should be considered when interpreting the results. First, although the dataset publisher and government source are identified, the measurement units are not specified in the supplied CSV. Second, the pollutant average has a right-skewed distribution and multiple high-value outliers, which can affect model errors. Third, pollutant_min and pollutant_max have strong correlations with pollutant_avg (0.84 and 0.85). Using these measurements to estimate the average can make the task easier and may create target-related information leakage depending on how the measurements are collected. These scores should not be interpreted as evidence of future forecasting ability.

Fourth, the classification labels are generated by splitting pollutant_avg at its median. The resulting classes represent lower and higher values relative to this dataset, not official AQI or health-risk categories. Fifth, the SVM visualization uses only latitude and longitude and reports training accuracy, so its score is not comparable with the held-out Logistic Regression and KNN scores.

Future work should document the original data source and measurement units, test the SVM using the same features and held-out split as the other classifiers, and use cross-validation for a more stable comparison. To investigate genuine future prediction, the task should be redesigned so that only information available before the prediction time is used. The project could also use verified AQI categories, if those labels and their official thresholds are available in the source data.

VII. CONCLUSION

This project developed a basic end-to-end workflow for cleaning, exploring, and modelling air-quality pollutant data. After removing rows with missing required pollutant values and checking for duplicates, 3,372 records and 11 columns remained. EDA showed that pollutant_avg is right-skewed, contains high-value outliers, and is strongly correlated with pollutant_min and pollutant_max. Polynomial Regression achieved the best regression result among the two evaluated models, with an R2 of 0.9564. Logistic Regression obtained 95.85% test accuracy and KNN obtained 95.41% test accuracy. The SVM figure provides an initial decision-boundary visualization, but its 55.40% training accuracy requires a fair held-out evaluation before comparison.

Overall, the project demonstrates how preprocessing, EDA, and multiple machine-learning methods can be combined to examine pollutant measurements. The strong relationship between the input minimum and maximum values and the average target must be considered when explaining the high regression scores. More complete dataset metadata and a consistent evaluation design would strengthen the analysis.

REFERENCES

[1] Central Pollution Control Board (CPCB), Ministry of Environment, Forest and Climate Change, "Real time Air Quality Index from various locations," Open Government Data (OGD) Platform India, CSV dataset. CSV resource: Real time Air Quality Index from various locations. Available: https://www.data.gov.in/catalog/real-time-air-quality-index. Accessed: 31 Oct. 2026.